

Almost exponential decay of quantum resonance states and Paley-Wiener type estimates in Gevrey spaces

Abstract

Let H_0 be a self-adjoint operator in some Hilbert space, and let λ_0 be a (possibly degenerate) eigenvalue of H_0 embedded in its essential spectrum $\sigma_{\text{ess}}(H_0)$ with corresponding eigenprojection Π_0 . For small $|\kappa|$, let $H(\kappa)$ be a family of perturbed Hamiltonians, which is analytic in a generalized Balslev-Combes sense. The corrections to exponential decay in

$$\Pi_0 e^{-itH(\kappa)} g(H(\kappa)) \Pi_0 = D(\kappa) e^{-ith(\kappa)} D(\kappa) + R(\kappa, t)$$

will be discussed, where $D(\kappa) = \Pi_0 + O(\kappa^2)$ ($\kappa \rightarrow 0$) and $h(\kappa)$ is some family of in general non self-adjoint bounded operators with $\text{Ran } h(\kappa) = \text{Ran } \Pi_0$, leaving $\text{Ran } \Pi_0$ invariant, and $0 \leq g \leq 1$ is a cut-off function with $g(\lambda_0) = 1$ and sufficiently small support. The main result is a sharp estimate of the remainder $R(\kappa, t)$ in terms of the Gevrey index $a > 1, b > 0$ of $g \in \Gamma^{a,b}(\mathbb{R})$:

$$\| R(\kappa, t) \| \leq O(\kappa^2) e^{-Ct^{1/a}} \quad (t \geq 0, C < ab^{-1/a}, \kappa \rightarrow 0).$$

This is joint work with M. Klein.

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